


Problem 2

$$\begin{aligned} (a) \quad E[\bar{Y} | X's] &= E\left[\frac{1}{n} \sum Y_i | X's\right] \\ &= \frac{1}{n} \sum E[Y_i | X's] \quad \text{by linearity } E[\cdot | X's] \\ &= \frac{1}{n} \sum E[Y_i | X_i] \quad \text{by strong exogeneity} \\ &= \frac{1}{n} \sum E[Y_1 | X_1] \quad \text{since we have an i.i.d. sample} \\ &= \frac{n}{n} E[Y_1 | X_1] \\ &= E[Y_1 | X_1] \end{aligned}$$

Remember i.i.d. $E[Y_i | X_i] = E[Y_j | X_j]$ for all $i=1, \dots, n$

We are, in Lecture 3, looking

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon$$

- $E[\varepsilon | X_i] = 0$

- $\text{Var}(\varepsilon | X_i) = \sigma^2$

Homoscedasticity is saying
conditional variance is fixed and constant
across all i .

$Y_i =$ GPA

$X_i =$ hours you study

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\text{Var}(Y_i | X_i) = \text{Var}(\varepsilon_i | X_i) = \sigma^2$$



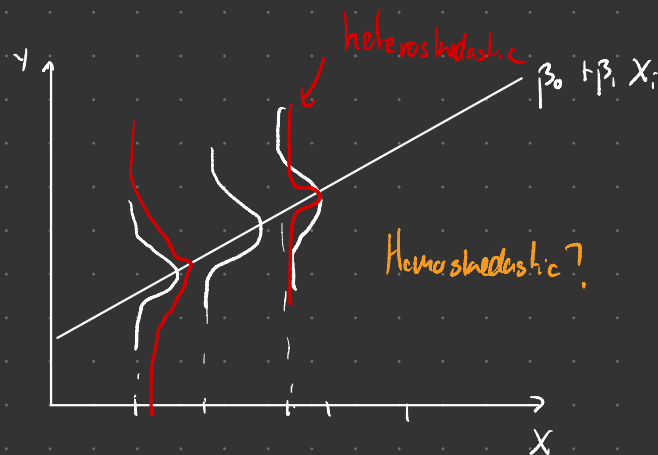
Under the conditional approach

we can derive some things

with this assumption:

$$\bullet \text{Var}(\hat{\beta} | X\text{'s})$$

Assuming (A5) $\varepsilon_i \sim \text{Normal}(0, \sigma^2)$



$$\hat{\beta} \sim \text{Normal}(\beta, \text{Var}(\hat{\beta} | X's))$$

$$\text{Var}(e_i | X_i) = \text{Var}(Y_i | X_i) = f(X_i)$$

$\hat{\beta}_0, \hat{\beta}_1$ OLS estimators are Random Variables

↓ Computed from a random sample, ↗

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$$

$$\text{Var}(\hat{\beta}_0 | X's)$$

$$\text{Var}(\hat{\beta}_1 | X's)$$

$$\text{Cor}(\hat{\beta}_0, \hat{\beta}_1 | X's)$$

$$\text{Var}(\hat{\underline{\beta}} | X's) = \begin{bmatrix} \text{Var}(\hat{\beta}_0 | X's) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's) \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's) & \text{Var}(\hat{\beta}_1 | X's) \end{bmatrix}$$

$$\text{Cov}(Y_i, Y_j | X\text{'s})$$

If you were to compute this you need $f(Y_i | X_i)$ by strong assumption.

$$\text{If } i=j \quad \text{Cov}(Y_i, Y_i | X\text{'s}) = \text{Var}(Y_i | X\text{'s}) = \text{Var}(Y_i | X_i) = \text{Var}(Y_1 | X_1)$$

$$\text{If } i \neq j \quad \text{Cov}(Y_i, Y_j | X\text{'s}) = 0$$

$$f(X_1, \dots, X_n, Y_1, \dots, Y_n) = f(X_1, Y_1) \times \dots \times f(X_n, Y_n)$$

→ You need $f(Y_i, Y_j | X_1, \dots, X_n) = f(Y_i | X_i) \cdot f(Y_j | X_j)$

Why can you write the above as

$$\text{Cov}(Y_i, Y_j | X\text{'s}) = I_{\{i=j\}} \text{Var}(Y_i | X_i) ?$$